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Large language models

Weekend 1: Language models with the Ge'ez games

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 16 October 2026 ፩፻፳፮

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Welcome, everyone

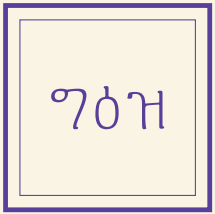
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Large language models

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We start our first module with an understanding of large language models



Introduction

Ge’ez

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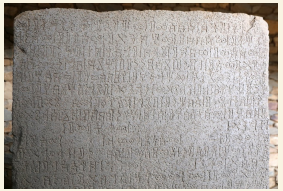
Introduction

Ge’ez

To make things more vivid, I’m going to use the language “Ge’ez” as our illustrative example

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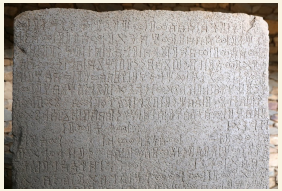




What is Ge’ez? Ge’ez is an ancient South Semitic language from the Horn of Africa

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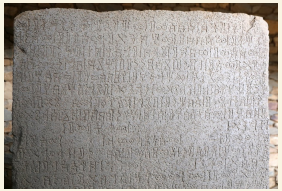
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originating in historical Abyssinia, today Ethiopia and Eritrea.

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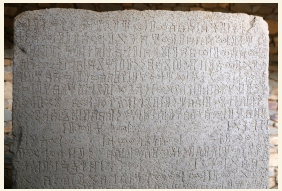




It is no longer a community's everyday spoken language

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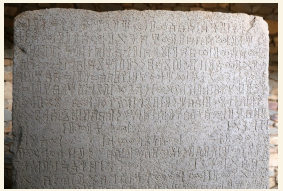




but it lives on as a liturgical language in several churches of the region

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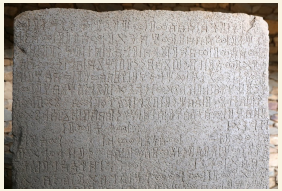




Ge’ez uses its own writing system, an abugida often called the “fidel,”

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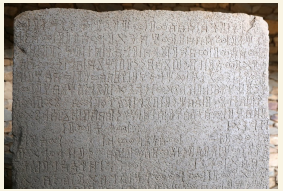
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where each sign represents a consonant with a built-in vowel

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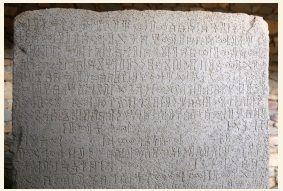
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Classical texts in Ge’ez include religious literature

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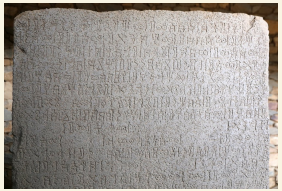
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royal chronicles, and translations of major works

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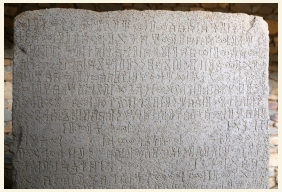
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giving us a window into the history of the region

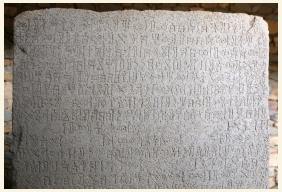
The Ge’ez language

- Ancient South Semitic language. The language originates from Abyssinia, what is now Ethiopia and Eritrea.



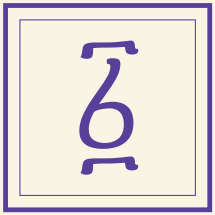
The Ge’ez language

- Ancient South Semitic language. The language originates from Abyssinia, what is now Ethiopia and Eritrea.
- We use this language as basis to study and investigate modern large language models.



The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 1

Structure

Game 1: objective

- We will show you some sentences in Ge'ez. They are related to either **city entities (city)** or **rural entities (countryside)**.

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Countryside	City
ʿEmäk lay ʾaḥar may	*bet nawīd lay ʾaḥar ʿIqlēsīya*
gāzā lay ʾasmāl may	*maytān lay ʾasmāl ʿIqlēsīya*
may lay ʾaḥar gāzā	*ʿIqlēsīya lay ʾaḥar maytān*
ṣadäq lay ʾaḥar ʿEmäk	*bet masḥāf lay ʾaḥar bet nawīd*
ṣadäq lay ʾaḥar gāzā	*bet masḥāf lay ʾaḥar maytān*

Insight

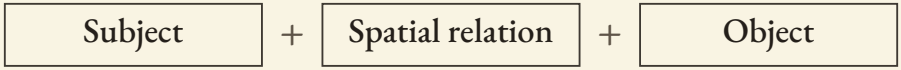
- We do not need to understand the meaning of a language nor its components to be able to extract information from it.

Insight

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- We are dealing with a very simple language:

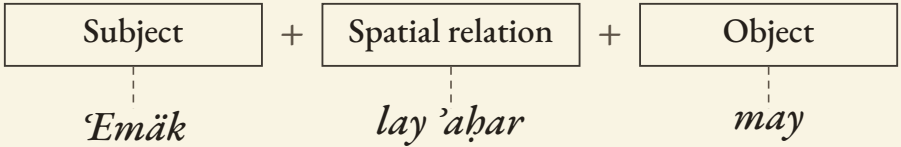
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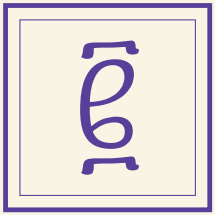
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Section 2

Imitation

Game 2: objective

- After observing sufficiently many examples of Ge'ez sentences you are now capable of producing not only syntactically valid, but also semantically meaningful sentences! Give it a try.

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- For example:

Game 2: objective

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- For example:
  - "Carlos is at ETH" is **semantically valid!**

Game 2: objective

- After observing sufficiently many examples of Ge'ez sentences you are now capable of producing not only syntactically valid, but also semantically meaningful sentences! Give it a try.
- For example:
  - "Carlos is at ETH" is **semantically valid!**
  - "ETH is at Carlos" is **not semantically valid!**

Insight

We don’t need to understand a language in order to produce meaningful sentences!

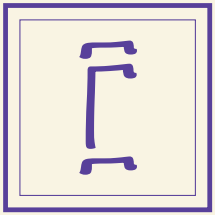
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Section 3

Prediction

Prediction

	ዕድል Probability	Language models are **probabilistic trees**.
	ሰንጠረዥ Table	**Probabilistic trees** can be represented with **tables**.
	ሥልጠና Learning	**Tables** can be learned from language texts.
	የውይይት ሮቦት Chatbot	The **table** can be used as a **chatbot**.

Game 3: Prediction

Alice plays tennis

Game 3: Prediction

Alice plays tennis

Alice plays golf

Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

Bob plays tennis

Game 3: Prediction

Alice plays tennis

Alice plays golf

Alice sings

Bob plays tennis

Bob plays chess

Probability trees

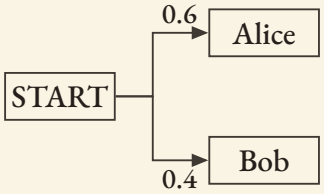
Alice plays tennis	Alice plays golf	Alice sings	Bob plays tennis	Bob plays chess

START

We can organize all these sentences into a probability tree!

Probability trees

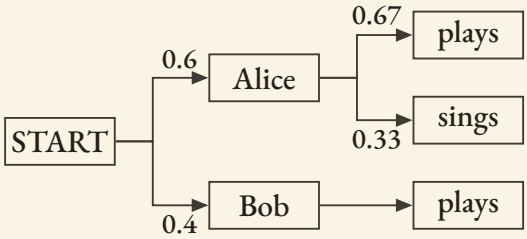
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We can organize all these sentences into a probability tree!

Probability trees

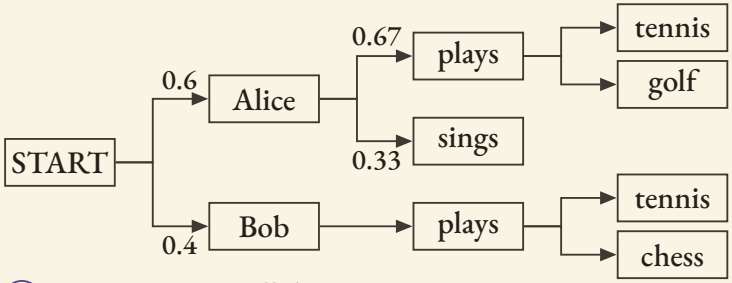
Alice plays tennis	Alice plays golf	Alice sings	Bob plays tennis	Bob plays chess



We can organize all these sentences into a probability tree!

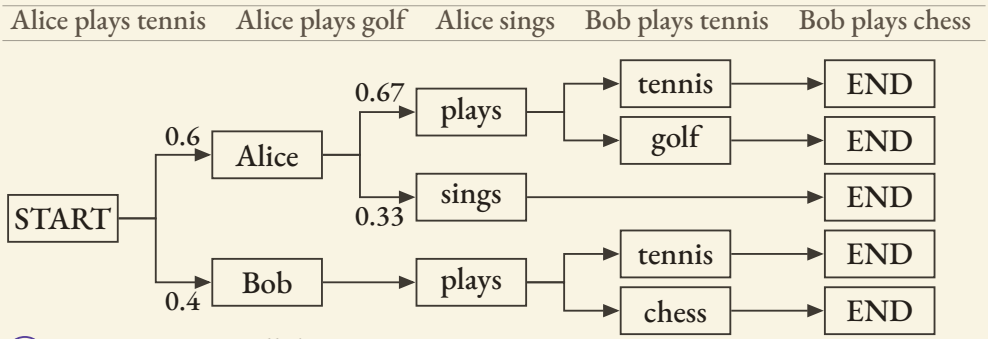
Probability trees

Alice plays tennis	Alice plays golf	Alice sings	Bob plays tennis	Bob plays chess



We can organize all these sentences into a probability tree!

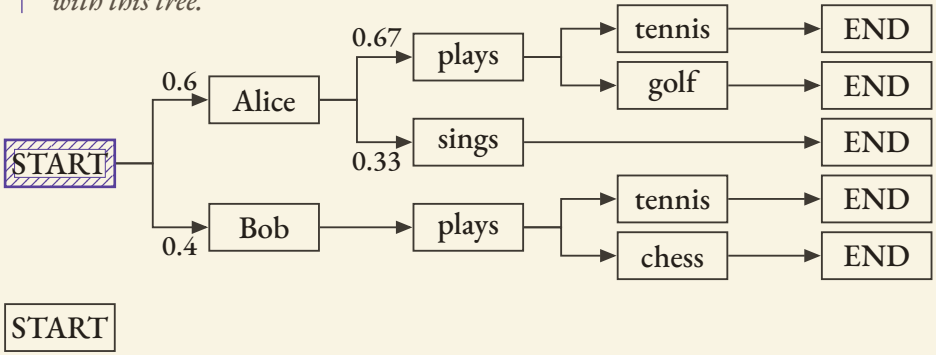
Probability trees



We can organize all these sentences into a probability tree!

Text generation as sampling

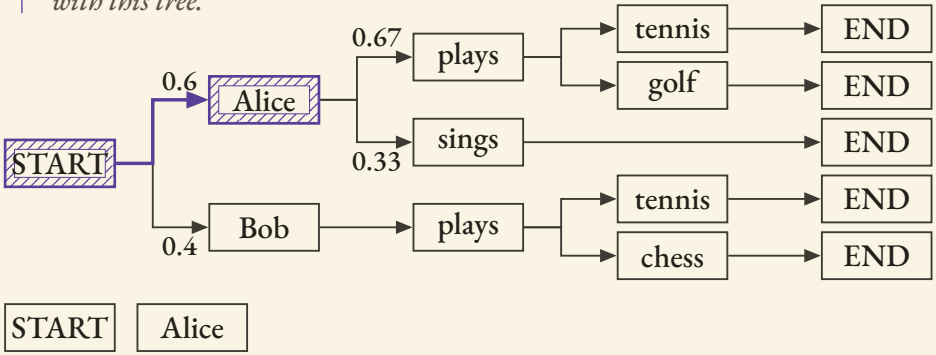
We can now produce sentences with this tree.



START

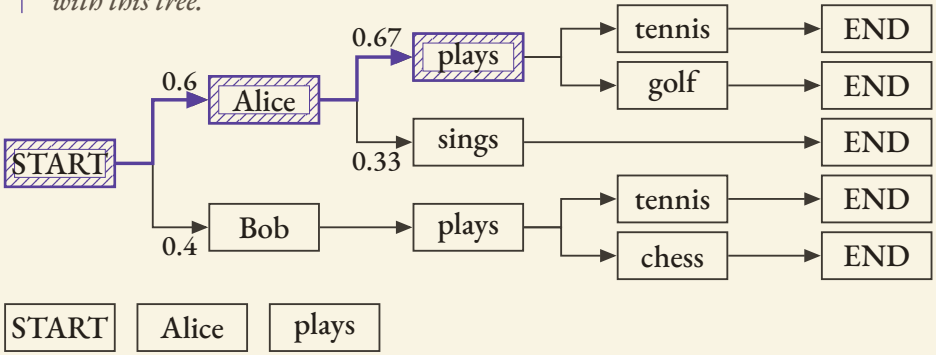
Text generation as sampling

We can now produce sentences with this tree.



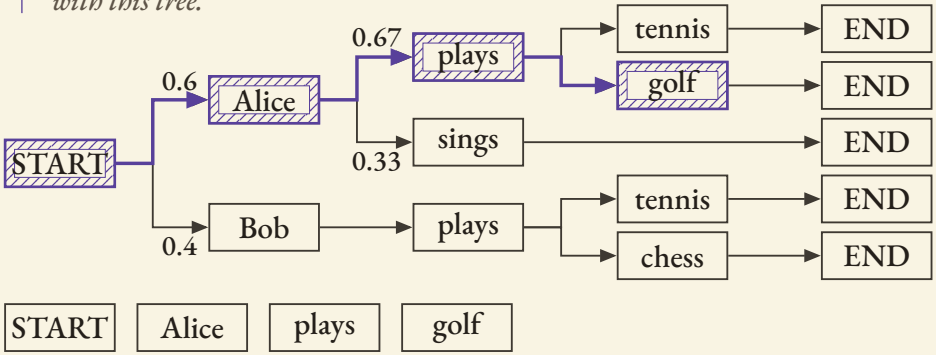
Text generation as sampling

We can now produce sentences with this tree.



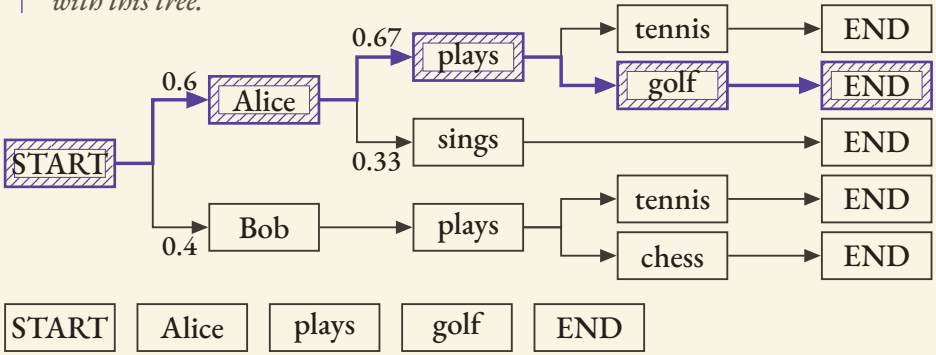
Text generation as sampling

We can now produce sentences with this tree.



Text generation as sampling

We can now produce sentences with this tree.



Prediction

	ዕድል Probability	Language models are **probabilistic trees**.
	ሰንጠረዥ Table	**Probabilistic trees** can be represented with **tables**.
	ሥልጠና Learning	**Tables** can be learned from language texts.
	የውይይት ሮቦት Chatbot	The **table** can be used as a **chatbot**.

Insight: We can represent a language model with a table!

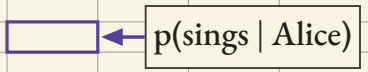
Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START								
Alice								
Bob								
Alice plays								
Alice sings								
Bob plays								
Alice plays golf								
Alice plays tennis								
Bob plays chess								
Bob plays tennis								



** We omit the keyword "START" in some prefixes for the sake of conciseness.*

Insight: We can represent a language model with a table!

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START								
Alice								
Bob								
Alice plays								
Alice sings								
Bob plays								
Alice plays golf								
Alice plays tennis								
Bob plays chess								
Bob plays tennis								



Insight: We can represent a language model with a table!

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START								
Alice								
Bob								
Alice plays								
Alice sings								
Bob plays								
Alice plays golf								
Alice plays tennis								
Bob plays chess								
Bob plays tennis								

p(golf | Bob plays)

Insight: We can represent a language model with a table!

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START								
Alice								
Bob								
Alice plays								
Alice sings								
Bob plays								
Alice plays golf								
Alice plays tennis								
Bob plays chess								
Bob plays tennis								

p(plays | Bob plays tennis)

Insight: We can represent a language model with a table!

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

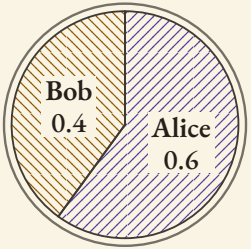
Sampling: spin a roulette

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						

Sampling: spin a roulette

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						

Each slice is as big as the chance of its word. A word with no chance gets no slice.

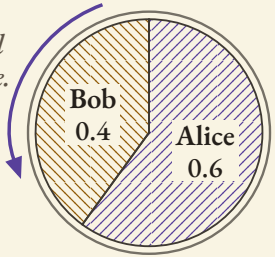


Sampling: spin a roulette

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						

Each slice is as big as the chance of its word. A word with no chance gets no slice.

Spin the roulette.

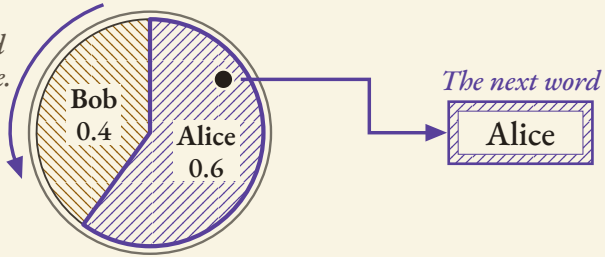


Sampling: spin a roulette

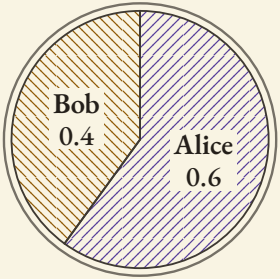
Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						

Each slice is as big as the chance of its word. A word with no chance gets no slice.

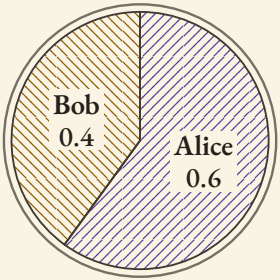
Spin the roulette.



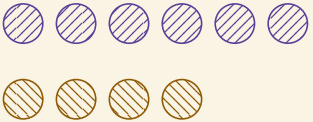
A bigger slice comes out more often



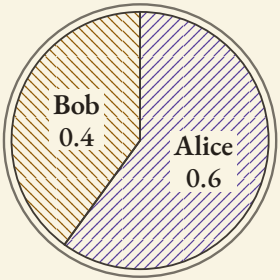
A bigger slice comes out more often



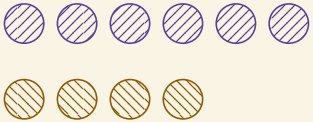
Ten spins, on average



A bigger slice comes out more often



Ten spins, on average



A real run of ten spins can differ a little. The more spins, the closer it gets.

We can now generate sentences by iterative sampling.

START

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

We can now generate sentences by iterative sampling.

START Alice

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

We can now generate sentences by iterative sampling.

START Alice plays

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

We can now generate sentences by iterative sampling.

START Alice plays golf

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

We can now generate sentences by iterative sampling.

START Alice plays golf END

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

Prediction



ዕድል
Probability

Language models are **probabilistic trees**.



ሰንጠረዥ
Table

Probabilistic trees can be represented with **tables**.



ሥልጠና
Learning

Tables can be learned from language texts.



የውይይት ሮቦት
Chatbot

The **table** can be used as a **chatbot**.

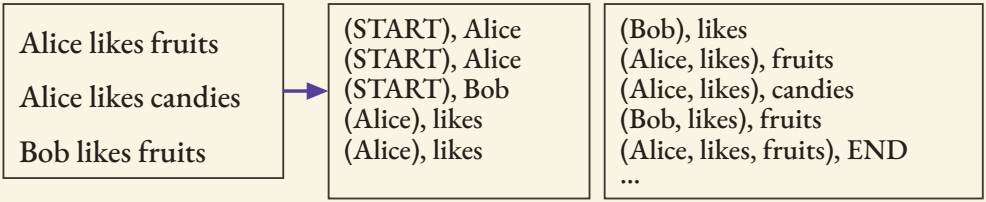
Training an autoregressive language model

Alice likes fruits

Alice likes candies

Bob likes fruits

Training an autoregressive language model



Training an autoregressive language model

Alice likes fruits

Alice likes candies

Bob likes fruits

(START), Alice
(START), Alice
(START), Bob
(Alice), likes
(Alice), likes

(Bob), likes
(Alice, likes), fruits
(Alice, likes), candies
(Bob, likes), fruits
(Alice, likes, fruits), END
...

Prefix	Alice	Bob	likes	candies	fruits	END
START	2/3	1/3				
Alice			1			
Bob			1			
Alice likes				1/2	1/2	
Bob likes					1	

Prediction

	ዕድል Probability	Language models are **probabilistic trees**.
	ሰንጠረዥ Table	**Probabilistic trees** can be represented with **tables**.
	ሥልጠና Learning	**Tables** can be learned from language texts.
	የውይይት ሮቦት Chatbot	The **table** can be used as a **chatbot**.

Your first chatbot!

What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

What does Alice like? Alice sings

What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess

Your first chatbot!

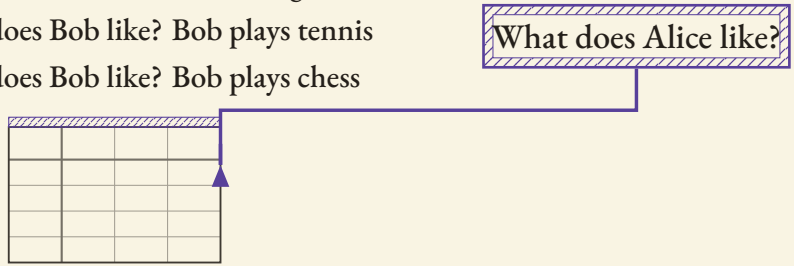
What does Alice like? Alice plays tennis

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What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess



Your first chatbot!

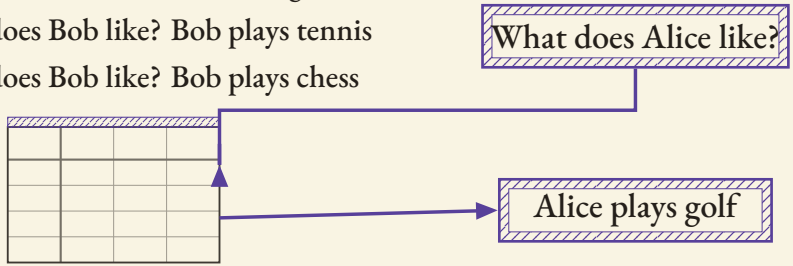
What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

What does Alice like? Alice sings

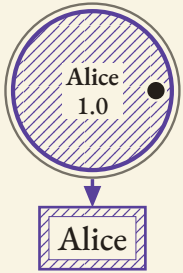
What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess



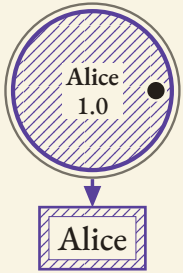
A chatbot answers one word at a time

What does Alice like?

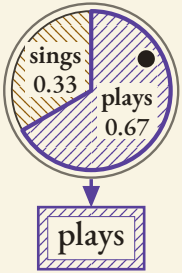


A chatbot answers one word at a time

What does Alice like?

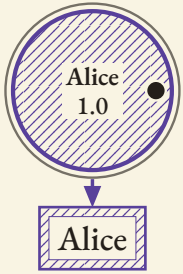


What does Alice like? Alice

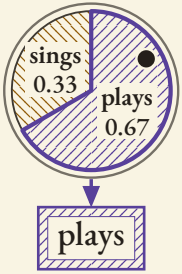


A chatbot answers one word at a time

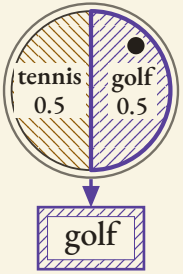
What does Alice like?



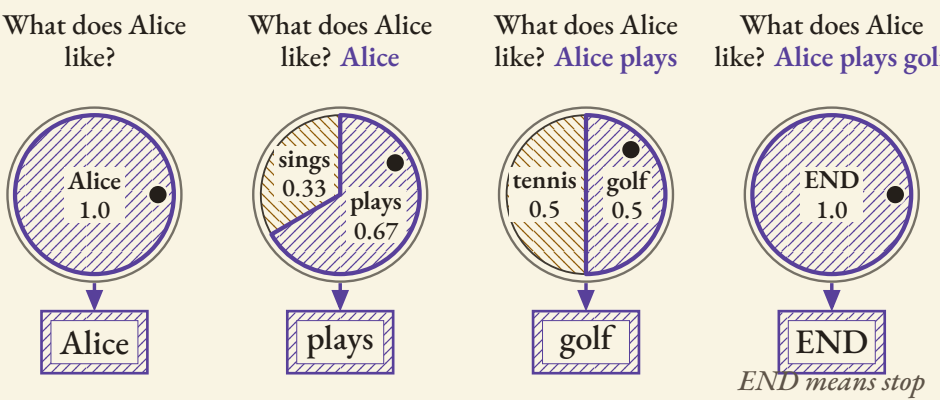
What does Alice like? Alice



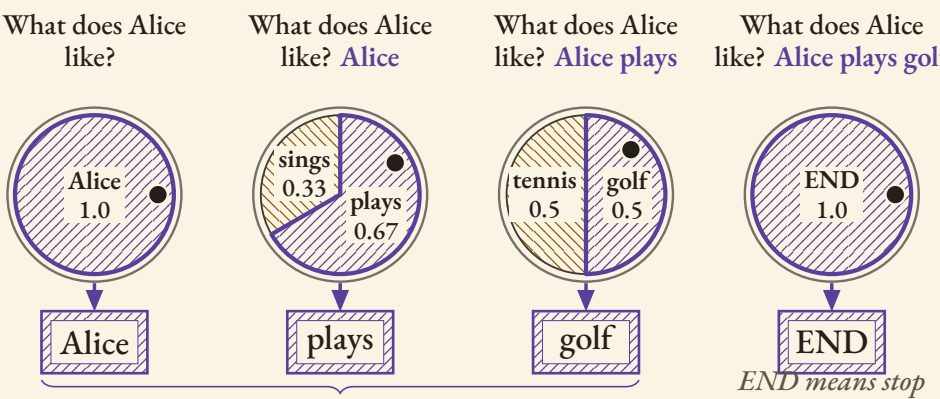
What does Alice like? Alice plays



A chatbot answers one word at a time



A chatbot answers one word at a time



Summary

- Language models are probabilistic trees.

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- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.

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- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.
- A **table** can be learned from a set of training sentences, by generating all combinations of prefix and continuation, and then calculating probabilities in the corresponding table.

Summary

- Language models are probabilistic trees.
- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.
- A **table** can be learned from a set of training sentences, by generating all combinations of prefix and continuation, and then calculating probabilities in the corresponding table.
- A conversation can then be carried out, by just taking the user's prompt, and using the **table** to generate **probable continuations**.

Prediction

	ዕድል Probability	Language models are **probabilistic trees**.
	ሰንጠረዥ Table	**Probabilistic trees** can be represented with **tables**.
	ሥልጠና Learning	**Tables** can be learned from language texts.
	የውይይት ሮቦት Chatbot	The **table** can be used as a **chatbot**.

The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.